

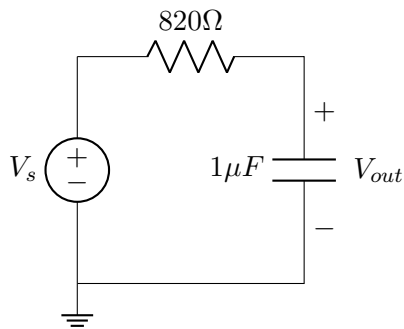
RC Circuit Response to Step, Pulse, and Square Inputs

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0.1 RC circuit diagram

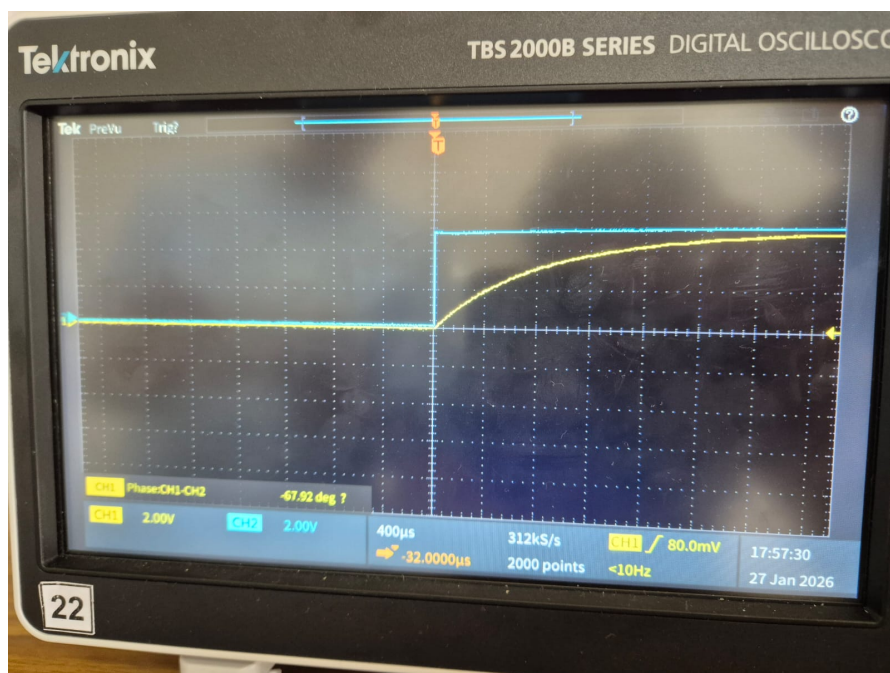


Here is the diagram of a RC circuit. RC circuits are low-pass filter circuits. The value of resistance and capacitance was chosen to be $R = 820\Omega$ and $C = 1\mu F$. We will now observe the behaviour and relation of V_{out} for different V_s . We will observe the response of different step, pulse and square inputs.

0.2 Step Input (Single Event)

0.2.1 Charging Response

Experimental Results



Signal Information

Blue Waveform (CH2): Step input (square rise edge). A step input of the form $V_s = 5u(t)$ was applied. This input signal could be generated by the AFG by applying a pulse signal of very very large width.

Yellow Waveform (CH1): Capacitor charging curve. This is the output and the voltage across the capacitor. We can see that this is the type of response we get for a step signal.

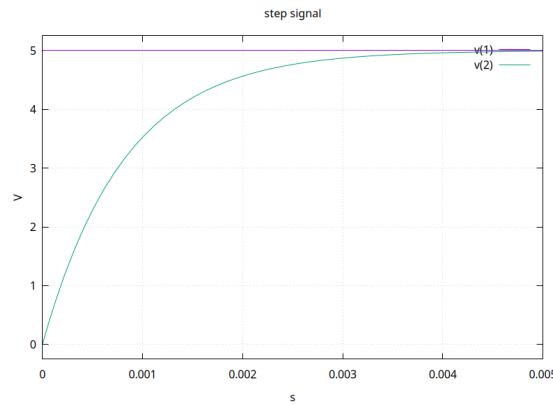
CH1 scale: 2.00 V/div

CH2 scale: 2.00 V/div

Time scale: 400 μs /division

Simulation

Following image is the simulation of the response of the RC circuit to the input waveform



Theoretical response

For a step input voltage $V_s(T)$: $V_C(t) = V_s (1 - e^{-t/RC})$

Computing time constant from measurement

For computing the time constant, we look at the charging waveform.

We compute time constant by measuring the time voltage across resistor takes to reach the voltage at one time constant.

Here $V_c(t)$ is the voltage across the capacitor at time t .

At one time constant $t = \tau = RC$:

$$V_C(\tau) = V_s(1 - e^{-1}) = 0.632V_s \quad (0.1)$$

$$V_C(\tau) = 0.632 \times 5 = 3.16V \quad (0.2)$$

From the graph we can take around 8 small divisions (1.6 divisions) to find the corresponding time. We find that the corresponding time is close to $800\mu s$ due to it being

around 2 time divisions away from the axis.

$$t \approx 800\mu s \quad (0.3)$$

This aligns with the actual value of resistances and capacitances we took. The theoretical value of t is $t = 820\mu s$. The error is less than 3%.

Verification at $t = 2\tau$

For a step input, the capacitor charging equation is:

$$V_C(t) = V_s \left(1 - e^{-t/\tau}\right) \quad (0.4)$$

At $t = 2\tau$:

$$V_C(2\tau) = V_s (1 - e^{-2}) \quad (0.5)$$

$$V_C(2\tau) = V_s (1 - 0.135) \quad (0.6)$$

$$V_C(2\tau) \approx 0.865 V_s = 0.865 \times 5V \quad (0.7)$$

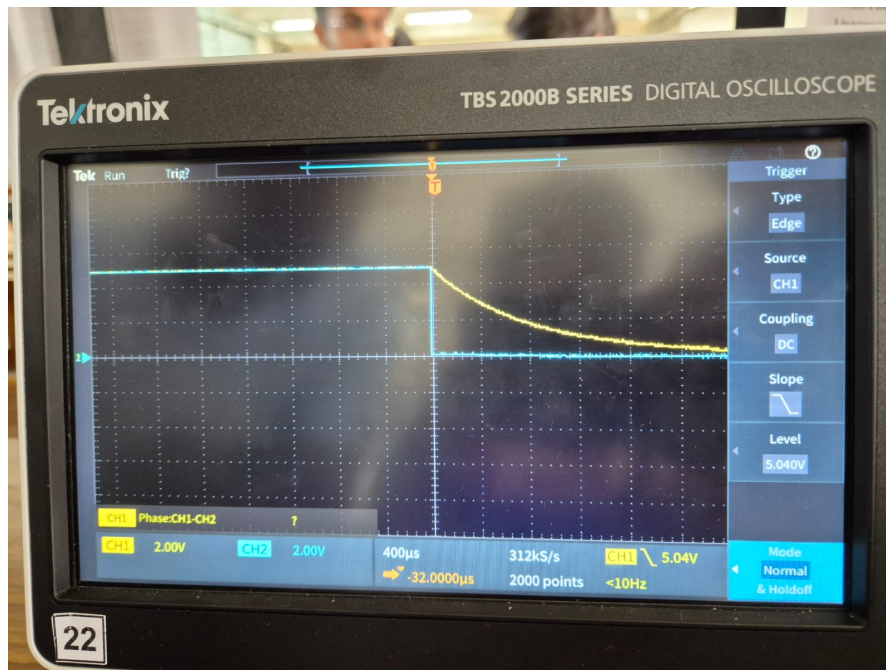
$$\boxed{V_C(2\tau) \approx 4.325V} \quad (0.8)$$

From the experimental waveform, the capacitor voltage at $t = 2\tau = 1.64ms$ (around 4 divisions in time scale) is observed to be close to $4.2V$, which is found to be 97.2% of the theoretical value.

Thus, the experimental result agrees well with the theoretical prediction, verifying the step response of the RC circuit.

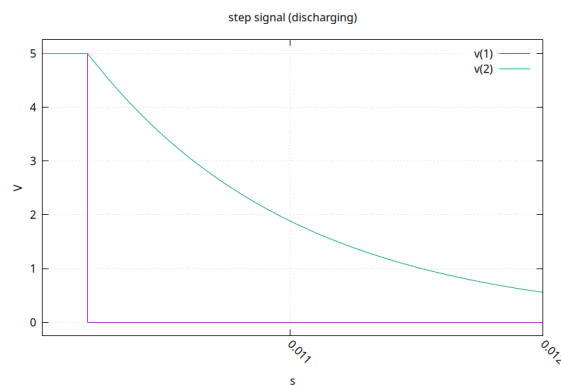
0.2.2 Discharging Response

Signal Information



Simulation

Following image is the simulation of the response of the RC circuit to the input waveform



When the input is removed: $V_C(t) = V_0 e^{-t/RC}$
At $t = \tau$:

$$V_C = 0.368V_0 = 0.368 \times 5V \quad (0.9)$$

$$= 1.84V \quad (0.10)$$

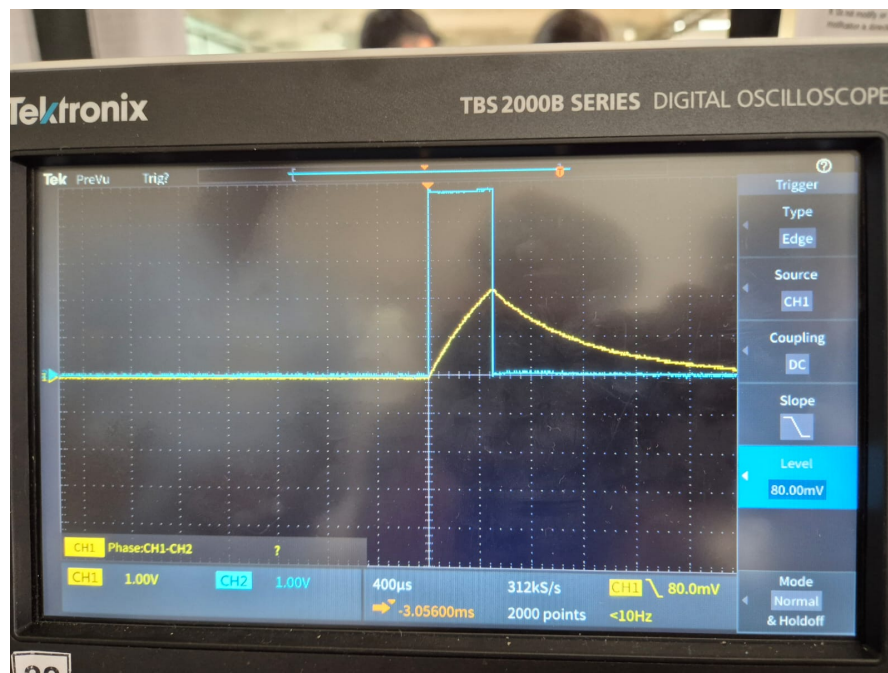
From the experimental waveform, the capacitor voltage at $t = \tau = 0.82\text{ ms}$ (around 2 divisions in time scale) is observed to be close to 1.8 V , which is found to be 97.82% of the theoretical value.

Thus, the experimental result agrees well with the theoretical prediction, verifying the step response of the RC circuit.

0.3 Pulse Input (Single Event)

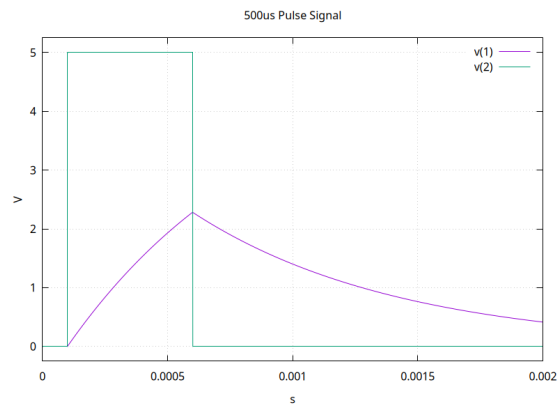
0.3.1 Pulse ($T = 200\mu\text{s}$)

Signal Information



Simulation

Following image is the simulation of the response of the RC circuit to the input waveform



Hand Calculation

Peak capacitor voltage:

$$V_{max} = V_s \left(1 - e^{-T/RC}\right) \quad (0.11)$$

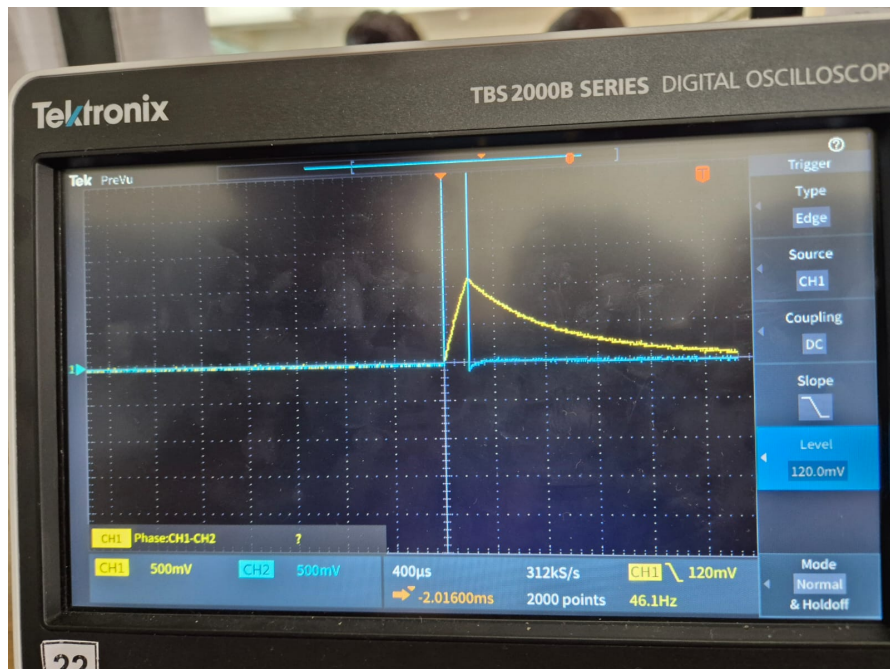
For $T = 200\mu s$:

$$V_{max} = 5 \times (1 - e^{-200/820})V \quad (0.12)$$

$$V_{max} = 1.08V \quad (0.13)$$

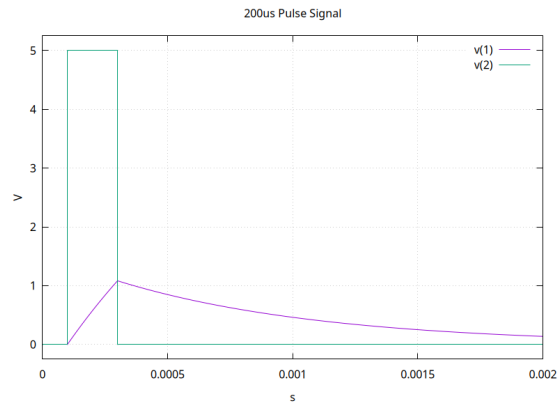
0.3.2 Pulse width ($T = 500\mu s$)

Signal Information



Simulation

Following image is the simulation of the response of the RC circuit to the input waveform



Hand Calculation

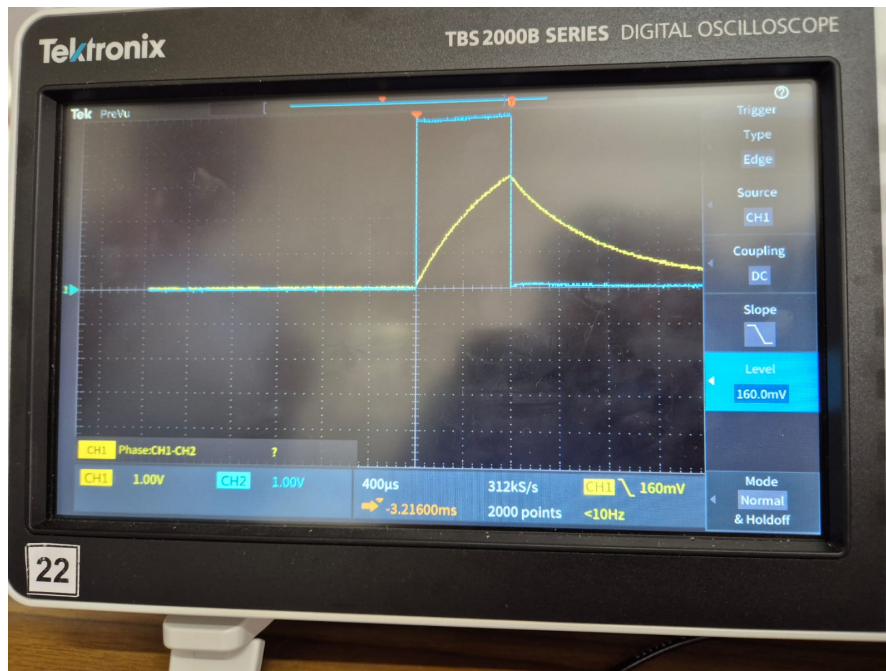
From (0.11), it follows that For $T = 500\mu s$:

$$V_{max} = 5 \times (1 - e^{-500/820})V \quad (0.14)$$

$$V_{max} = 2.28V \quad (0.15)$$

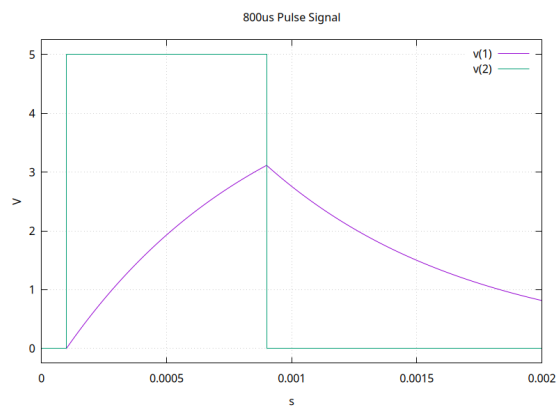
0.3.3 Pulse $T = 800\mu s$

Signal Information



Simulation

Following image is the simulation of the response of the RC circuit to the input waveform



Hand Calculation

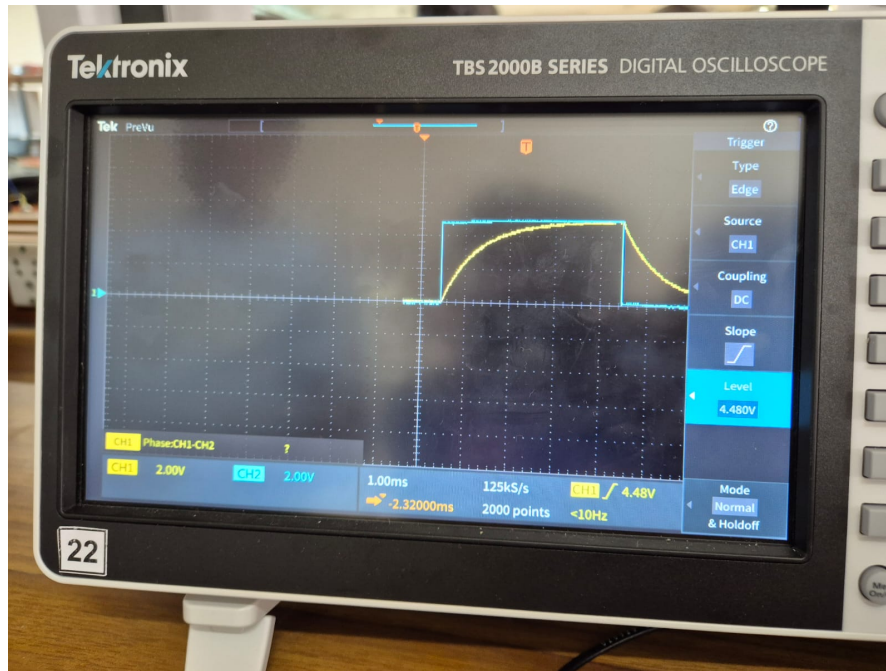
From (0.11), it follows that

$$V_{max} = 5 \times (1 - e^{-800/820})V \quad (0.16)$$

$$V_{max} = 3.12V \quad (0.17)$$

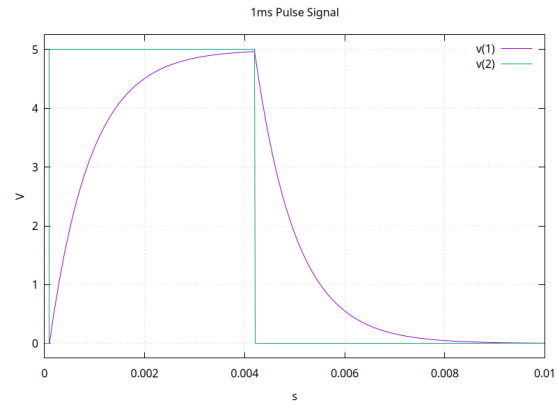
0.3.4 Pulse ($T = 1ms$)

Signal Information



Simulation

Following image is the simulation of the response of the RC circuit to the input waveform



Hand Calculation

From (0.11), it follows that

$$V_{max} = 5 \times (1 - e^{-1000/820})V \quad (0.18)$$

$$V_{max} = 3.52V \quad (0.19)$$

0.4 Square-Wave Input (Transient and Steady-State)

0.4.1 $RC \gg T$

Signal Information

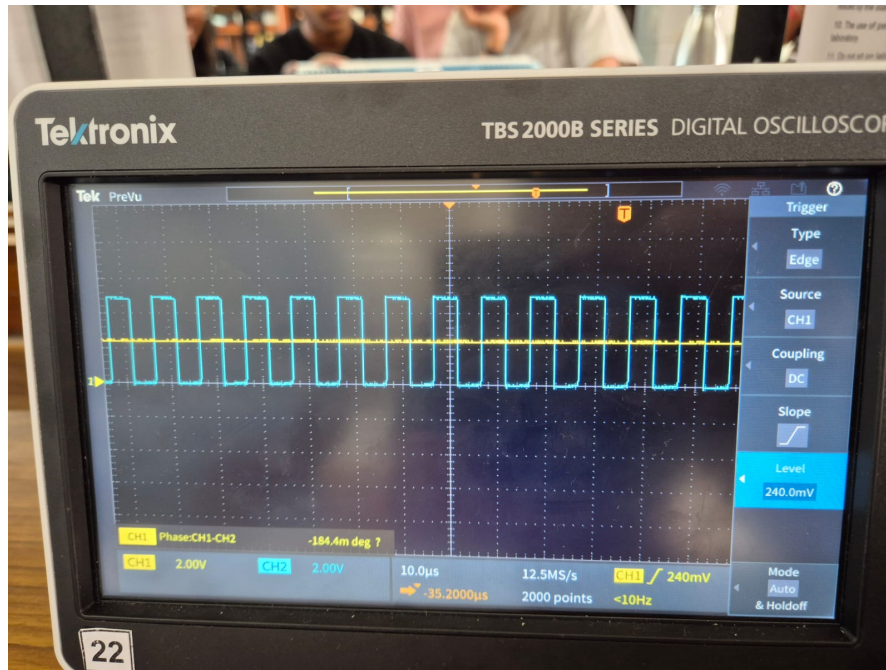
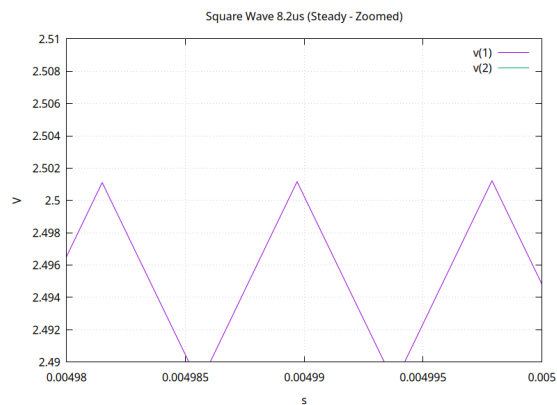


Figure 1: Steady-State Response

Simulation

Following image is the simulation of the response of the RC circuit to the input waveform



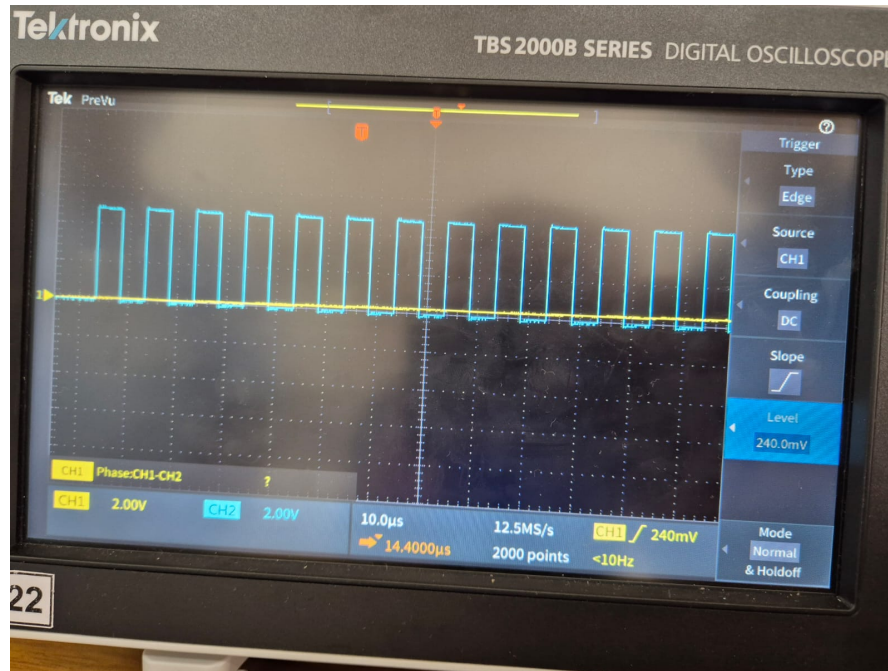
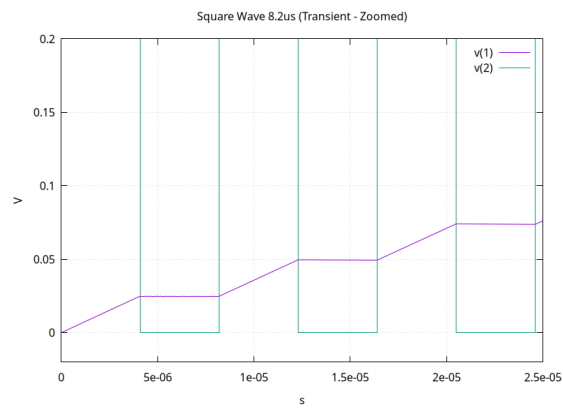


Figure 2: Transient Response

Simulation

Following image is the simulation of the response of the RC circuit to the input waveform



Hand Calculations

For capacitor charging:

$$V_C(t) = V_i + (V_f - V_i) \left(1 - e^{-t/RC}\right) \quad (0.20)$$

Since $t \leq \frac{T}{2} \ll RC$,

$$\frac{t}{RC} \ll 1 \quad (0.21)$$

Using approximation:

$$e^{-t/RC} \approx 1 - \frac{t}{RC} \quad (0.22)$$

$$V_C(t) \approx V_i + (V_f - V_i) \frac{t}{RC} \quad (0.23)$$

Inference

The capacitor voltage varies almost linearly with time. Hence, the output waveform is approximately triangular.

RC circuit behaves as an integrator

(0.24)

0.4.2 Case 2: $RC \approx T$

Signal Information

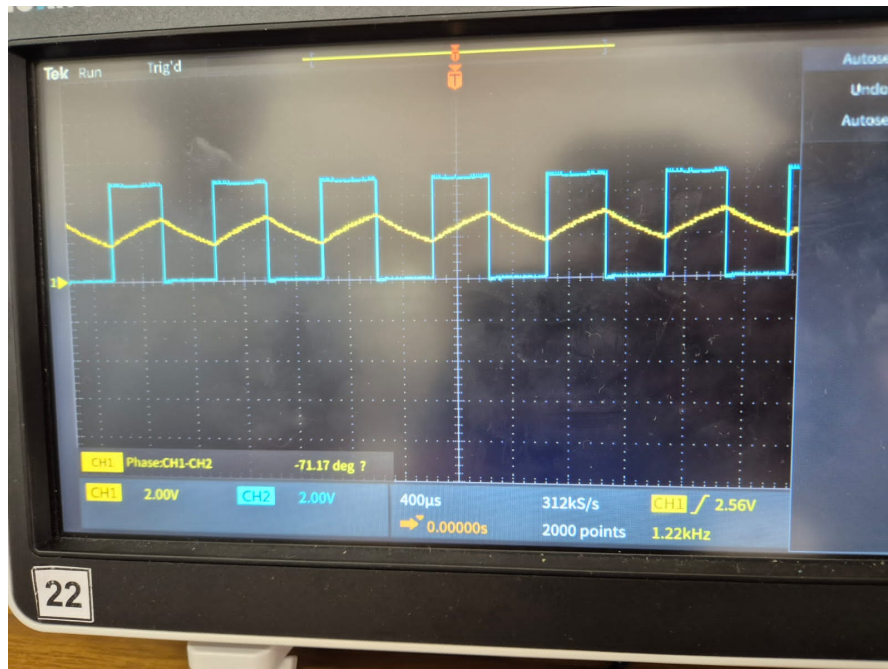


Figure 3: Steady-State Response

Simulation

Following image is the simulation of the response of the RC circuit to the input waveform

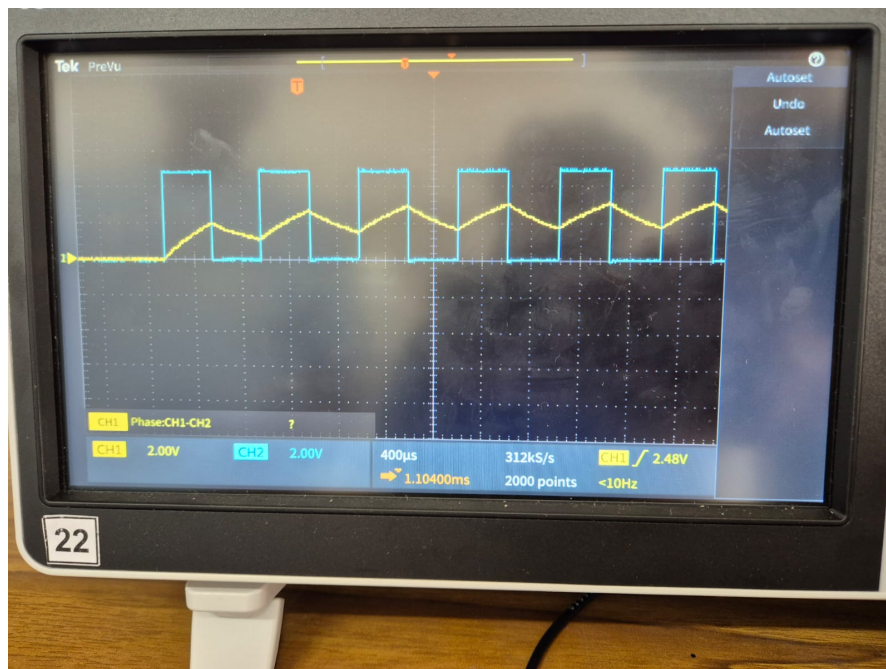
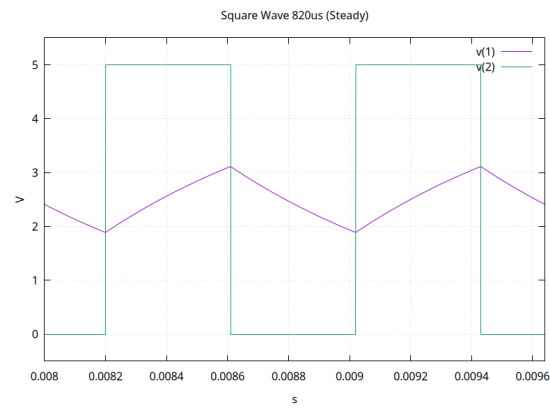
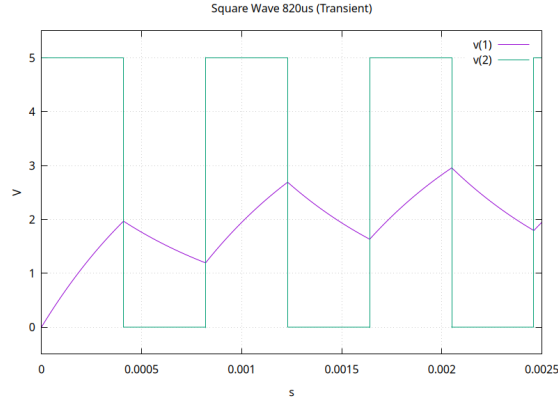


Figure 4: Transient Response

Simulation

Following image is the simulation of the response of the RC circuit to the input waveform



$$T \approx RC = 0.82 \text{ ms} \quad (0.25)$$

Hand Calculations

During the charging interval:

$$V_C(t) = V_s \left(1 - e^{-t/RC}\right) \quad (0.26)$$

At the end of half cycle:

$$t = \frac{T}{2} \approx \frac{RC}{2} \quad (0.27)$$

$$V_C = V_s (1 - e^{-0.5}) \quad (0.28)$$

$$V_C \approx 0.393V_s \quad (0.29)$$

During discharge:

$$V_C(t) = V_0 e^{-t/RC} \quad (0.30)$$

At $t = \frac{T}{2}$:

$$V_C = V_0 e^{-0.5} \approx 0.607V_0 \quad (0.31)$$

Inference

The capacitor partially charges and partially discharges in each cycle. Both transient and steady-state components are present.

$$\boxed{\text{Maximum transient behavior is observed}} \quad (0.32)$$

0.4.3 Case 3: $RC \ll T$

Signal Information

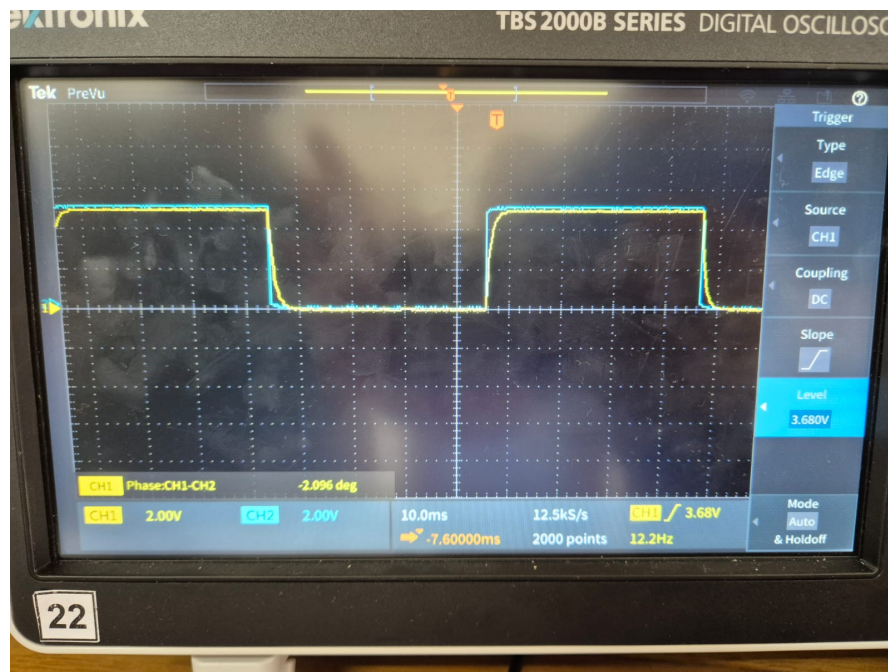


Figure 5: Steady-State Response

Simulation

Following image is the simulation of the response of the RC circuit to the input waveform

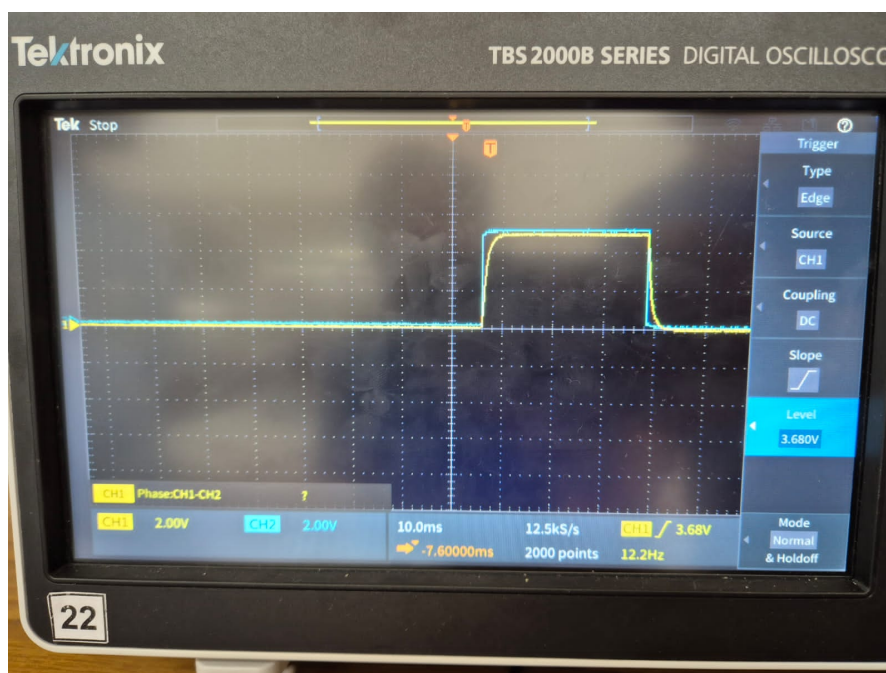
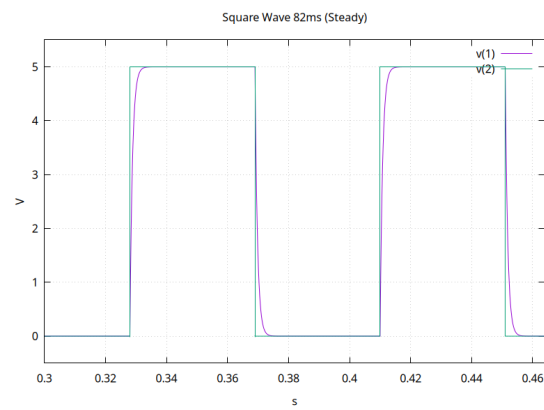
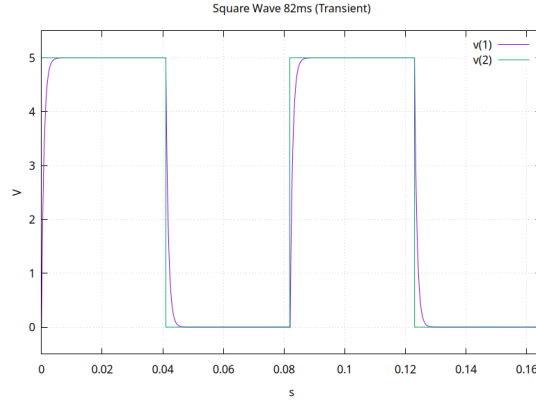


Figure 6: Transient Response

Simulation

Following image is the simulation of the response of the RC circuit to the input waveform



$$T \gg RC \quad (0.33)$$

$$T \gg 0.82 \text{ ms} \quad (0.34)$$

Hand Calculations

During ON time:

$$t = \frac{T}{2} \gg RC \quad (0.35)$$

$$e^{-t/RC} \approx 0 \quad (0.36)$$

$$V_C \approx V_s \quad (0.37)$$

During OFF time:

$$V_C(t) = V_s e^{-t/RC} \quad (0.38)$$

For $t \gg RC$:

$$V_C \approx 0 \quad (0.39)$$

Inference

The capacitor fully charges and discharges within each cycle.

Output closely follows the input waveform

(0.40)

0.5 Conclusion

All our experimental results are nearly accurate and within an error of 5% of theoretical results. The simulations have been generated using NGSpice and the code for generating all the simulations are available in our Github repository. The link for the repository is here

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https://github.com/Vivek-Kumar-1907/ee1200/tree/main/assignment2/simulations/
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